

Architecture Diagram Explanation

Overview

The Restaurant Customer Analytics Platform is built using a modern AWS Medallion Architecture consisting of Bronze, Silver, and Gold data layers. The solution enables daily customer analytics, business intelligence reporting, and executive decision-making.

Source and Ingestion Layer

The platform begins with SQL Server source tables containing restaurant transactional data.

AWS Database Migration Service (DMS) continuously extracts data using Change Data Capture (CDC), allowing incremental updates without requiring full reloads.

Benefits:

- Near real-time updates
 - Reduced load on source systems
 - Automatic retry and recovery
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Bronze Layer

The Bronze layer stores raw source data exactly as received.

Datasets:

- order_items
- order_item_options
- date_dim

Purpose:

- Data preservation
- Historical auditing
- Recovery and replay capability

All data is encrypted using AWS KMS and versioned in Amazon S3.

Data Validation

AWS Glue Data Quality validates incoming datasets.

Checks include:

- Missing values
- Null columns
- Duplicate records
- Schema changes
- Key integrity validation

Validation reports are written to a dedicated S3 audit location.

Silver Layer

The Silver layer contains cleansed and standardized business entities.

Examples:

- customer_orders
- customer_daily_metrics
- loyalty_metrics
- location_metrics

Data is stored in Parquet format and partitioned for performance optimization.

Purpose:

- Improved query performance
 - Standardized business logic
 - Reusable datasets
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Gold Layer

The Gold layer contains business-ready analytics datasets.

Examples:

- customer_ltv_daily
- customer_rfm
- customer_churn
- loyalty_impact
- location_performance

- sales_trends
- discount_analysis

Purpose:

- Executive reporting
 - Dashboard consumption
 - Self-service analytics
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Metadata and Query Layer

AWS Glue Catalog maintains metadata definitions for all datasets.

Amazon Athena provides serverless SQL access to Gold datasets.

Benefits:

- No infrastructure management
 - Fast analytical queries
 - Cost-efficient reporting
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Dashboard Layer

Streamlit provides interactive business dashboards.

Dashboard modules include:

- Customer Segmentation
- Customer Lifetime Value
- Churn Analysis
- Loyalty Impact
- Location Performance
- Sales Trends
- Discount Effectiveness

Users can filter and analyze data by:

- Date
 - Customer
 - Restaurant
 - Loyalty Status
 - Product Category
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Security and Operations

Security controls span the entire architecture.

Components:

- AWS KMS
- IAM Roles
- AWS Secrets Manager
- Amazon CloudWatch
- Amazon SNS

These services provide encryption, access control, monitoring, and alerting.

Business Value

This architecture enables the organization to:

- Understand customer behavior
- Improve customer retention
- Measure loyalty program effectiveness
- Optimize promotions
- Improve location performance
- Support data-driven decision making

The solution is scalable, secure, and fully aligned with AWS best practices while satisfying all project requirements.